```
import pandas as pd
import numpy as np

df = pd.read_csv('/content/covid_vaccine_statewise.csv')
df.head()
```

	Updated On	State	Total Doses Administered	Sessions	Sites	First Dose Administered	Second Dose Administered	Male (Doses Administered)	Female (Doses Administered)	Transgender (Doses Administered)	...	18-44 Years (Doses Administered)	45-60 Years (Doses Administered)	60+ Years (Doses Administered)	18-44 Years(Individuals Vaccinated)	45-60 Years(Individuals Vaccinated)	Years
0	16/01/2021	India	48276.0	3455.0	2957.0	48276.0	0.0	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
1	17/01/2021	India	58604.0	8532.0	4954.0	58604.0	0.0	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
2	18/01/2021	India	99449.0	13611.0	6583.0	99449.0	0.0	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
3	19/01/2021	India	195525.0	17855.0	7951.0	195525.0	0.0	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
4	20/01/2021	India	251280.0	25472.0	10504.0	251280.0	0.0	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	

5 rows × 24 columns

```
df.tail()
```

	Updated On	State	Total Doses Administered	Sessions	Sites	First Dose Administered	Second Dose Administered	Male (Doses Administered)	Female (Doses Administered)	Transgender (Doses Administered)	...	18-44 Years (Doses Administered)	45-60 Years (Doses Administered)	60+ Years (Doses Administered)	18-44 Years(Individuals Vaccinated)	45-60 Years(Individuals Vaccinated)	Yea
7840	11/08/2021	West Bengal	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
7841	12/08/2021	West Bengal	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
7842	13/08/2021	West Bengal	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
7843	14/08/2021	West Bengal	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	
7844	15/08/2021	West Bengal	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	

5 rows × 24 columns

```
df.dtypes
```

	0
Updated On	object
State	object
Total Doses Administered	float64
Sessions	float64
Sites	float64
First Dose Administered	float64
Second Dose Administered	float64
Male (Doses Administered)	float64
Female (Doses Administered)	float64
Transgender (Doses Administered)	float64
Covaxin (Doses Administered)	float64
CoviShield (Doses Administered)	float64
Sputnik V (Doses Administered)	float64
AEFI	float64
18-44 Years (Doses Administered)	float64
45-60 Years (Doses Administered)	float64
60+ Years (Doses Administered)	float64
18-44 Years(Individuals Vaccinated)	float64
45-60 Years(Individuals Vaccinated)	float64
60+ Years(Individuals Vaccinated)	float64
Male(Individuals Vaccinated)	float64
Female(Individuals Vaccinated)	float64
Transgender(Individuals Vaccinated)	float64
Total Individuals Vaccinated	float64

dtype: object

```
df.isnull().sum()
```

	0
Updated On	0
State	0
Total Doses Administered	224
Sessions	224
Sites	224
First Dose Administered	224
Second Dose Administered	224
Male (Doses Administered)	384
Female (Doses Administered)	384
Transgender (Doses Administered)	384
Covaxin (Doses Administered)	224
CoviShield (Doses Administered)	224
Sputnik V (Doses Administered)	4850
AEFI	2407
18-44 Years (Doses Administered)	6143
45-60 Years (Doses Administered)	6143
60+ Years (Doses Administered)	6143
18-44 Years(Individuals Vaccinated)	4112
45-60 Years(Individuals Vaccinated)	4111
60+ Years(Individuals Vaccinated)	4111
Male(Individuals Vaccinated)	7685
Female(Individuals Vaccinated)	7685
Transgender(Individuals Vaccinated)	7685
Total Individuals Vaccinated	1926

dtype: int64

```
for col in df.select_dtypes(include=np.number).columns:
    df[col] = df[col].fillna(0)

display(df.isnull().sum())
```

```
Null values after dropping high-null columns and imputing remaining numerical nulls with 0:
```

	0
Updated On	0
State	0
Total Doses Administered	0
Sessions	0
Sites	0
First Dose Administered	0
Second Dose Administered	0
Male (Doses Administered)	0
Female (Doses Administered)	0
Transgender (Doses Administered)	0
Covaxin (Doses Administered)	0
CoviShield (Doses Administered)	0
Sputnik V (Doses Administered)	0
AEFI	0
18-44 Years (Doses Administered)	0
45-60 Years (Doses Administered)	0
60+ Years (Doses Administered)	0
18-44 Years(Individuals Vaccinated)	0
45-60 Years(Individuals Vaccinated)	0
60+ Years(Individuals Vaccinated)	0
Male(Individuals Vaccinated)	0
Female(Individuals Vaccinated)	0
Transgender(Individuals Vaccinated)	0
Total Individuals Vaccinated	0

dtype: int64

```
print("DataFrame Information:")
df.info()
```

```
DataFrame Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7845 entries, 0 to 7844
Data columns (total 24 columns):
 #   Column                                  Non-Null Count  Dtype
---  ------                                  --------------  -----
 0   Updated On                              7845 non-null   object
 1   State                                   7845 non-null   object
 2   Total Doses Administered                7845 non-null   float64
 3   Sessions                                7845 non-null   float64
 4    Sites                                  7845 non-null   float64
 5   First Dose Administered                 7845 non-null   float64
 6   Second Dose Administered                7845 non-null   float64
 7   Male (Doses Administered)               7845 non-null   float64
 8   Female (Doses Administered)             7845 non-null   float64
 9   Transgender (Doses Administered)        7845 non-null   float64
 10   Covaxin (Doses Administered)           7845 non-null   float64
 11  CoviShield (Doses Administered)         7845 non-null   float64
 12  Sputnik V (Doses Administered)          7845 non-null   float64
 13  AEFI                                    7845 non-null   float64
 14  18-44 Years (Doses Administered)        7845 non-null   float64
 15  45-60 Years (Doses Administered)        7845 non-null   float64
 16  60+ Years (Doses Administered)          7845 non-null   float64
 17  18-44 Years(Individuals Vaccinated)     7845 non-null   float64
 18  45-60 Years(Individuals Vaccinated)     7845 non-null   float64
 19  60+ Years(Individuals Vaccinated)       7845 non-null   float64
 20  Male(Individuals Vaccinated)            7845 non-null   float64
 21  Female(Individuals Vaccinated)          7845 non-null   float64
 22  Transgender(Individuals Vaccinated)     7845 non-null   float64
 23  Total Individuals Vaccinated            7845 non-null   float64
dtypes: float64(22), object(2)
memory usage: 1.4+ MB
```

```
print("\nDataFrame Description (numerical columns):")
display(df.describe())
```

DataFrame Description (numerical columns):

	Total Doses Administered	Sessions	Sites	First Dose Administered	Second Dose Administered	Male (Doses Administered)	Female (Doses Administered)	Transgender (Doses Administered)	Covaxin (Doses Administered)	CoviShield (Doses Administered)	...	18-44 Years (Doses Administered)	45-60 Years (Doses Administered)	60+ Years (Doses Administered)	18-44 Years(Individuals Vaccinated)
count	7.845000e+03	7.845000e+03	7845.000000	7.845000e+03	7.845000e+03	7.845000e+03	7.845000e+03	7845.000000	7.845000e+03	7.845000e+03	...	7.845000e+03	7.845000e+03	7.845000e+03	7.845000e+03
mean	8.925819e+06	4.655521e+05	2217.688719	7.202710e+06	1.723109e+06	3.442955e+06	3.013328e+06	1106.052135	1.014841e+06	7.894514e+06	...	1.903541e+06	1.614603e+06	1.223966e+06	6.642287e+05
std	3.695473e+07	1.885710e+06	7181.404758	2.954714e+07	7.467353e+06	1.696664e+07	1.479335e+07	5789.789364	4.391675e+06	3.253793e+07	...	1.290784e+07	1.081032e+07	8.169012e+06	3.858230e+06
min	0.000000e+00	0.000000e+00	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000	0.000000e+00	0.000000e+00	...	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
25%	1.129780e+05	4.913000e+03	63.000000	9.479600e+04	8.724000e+03	4.078100e+04	3.478800e+04	4.000000	0.000000e+00	1.105880e+05	...	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
50%	7.445650e+05	3.634500e+04	541.000000	6.170910e+05	1.205330e+05	3.363550e+05	2.886440e+05	93.000000	6.745000e+03	6.821070e+05	...	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
75%	6.344469e+06	3.263560e+05	1642.000000	5.174960e+06	1.105948e+06	2.571915e+06	2.390020e+06	730.000000	7.164710e+05	5.737576e+06	...	0.000000e+00	0.000000e+00	0.000000e+00	2.585950e+05
max	5.132284e+08	3.501031e+07	73933.000000	4.001504e+08	1.130780e+08	2.701636e+08	2.395186e+08	98275.000000	6.236742e+07	4.468251e+08	...	2.243304e+08	1.667575e+08	1.186927e+08	9.224315e+07

8 rows × 22 columns

```
# b. Number of persons state wise vaccinated for first dose in India
first_dose_statewise = df.groupby('State')['First Dose Administered'].sum().sort_values(ascending=False)
print("\nTotal First Doses Administered State-wise:")
display(first_dose_statewise)
```

Total First Doses Administered State-wise:

State	First Dose Administered
India	2.826214e+10
Uttar Pradesh	2.788411e+09
Maharashtra	2.784364e+09
Rajasthan	2.201044e+09
Gujarat	2.131646e+09
Karnataka	1.873330e+09
Madhya Pradesh	1.796605e+09
West Bengal	1.796450e+09
Bihar	1.470503e+09
Tamil Nadu	1.288533e+09
Andhra Pradesh	1.232861e+09
Kerala	1.193845e+09
Odisha	1.032633e+09
Telangana	8.803206e+08
Chhattisgarh	7.960029e+08
Haryana	7.557984e+08
Delhi	6.243395e+08
Jharkhand	6.036737e+08
Assam	5.856002e+08
Punjab	5.843466e+08
Jammu and Kashmir	4.101018e+08
Uttarakhand	3.631914e+08
Himachal Pradesh	3.162940e+08
Tripura	1.926897e+08
Goa	7.599137e+07
Manipur	6.740957e+07
Meghalaya	6.261597e+07
Arunachal Pradesh	4.900498e+07
Mizoram	4.787308e+07
Chandigarh	4.470310e+07
Nagaland	4.241077e+07
Puducherry	4.134686e+07
Sikkim	3.698093e+07
Dadra and Nagar Haveli and Daman and Diu	3.359506e+07
Ladakh	1.780925e+07
Andaman and Nicobar Islands	1.642585e+07
Lakshadweep	4.363655e+06

dtype: float64

```
# c. Number of persons state wise vaccinated for second dose in India
second_dose_statewise = df.groupby('State')['Second Dose Administered'].sum().sort_values(ascending=False)
print("\nTotal Second Doses Administered State-wise:")
display(second_dose_statewise)
```

```
Total Second Doses Administered State-wise:
```

State	Second Dose Administered
India	6.759621e+09
Maharashtra	7.128811e+08
Gujarat	6.004184e+08
West Bengal	5.861469e+08
Uttar Pradesh	5.544351e+08
Rajasthan	4.917030e+08
Karnataka	4.271872e+08
Kerala	3.640488e+08
Andhra Pradesh	3.588176e+08
Madhya Pradesh	3.169330e+08
Tamil Nadu	2.906706e+08
Bihar	2.707906e+08
Odisha	2.513028e+08
Telangana	1.981529e+08
Delhi	1.882189e+08
Chhattisgarh	1.721204e+08
Haryana	1.586561e+08
Assam	1.307888e+08
Jharkhand	1.221211e+08
Punjab	1.211210e+08
Himachal Pradesh	7.383858e+07
Tripura	6.527014e+07
Arunachal Pradesh	1.193232e+07
Manipur	1.185815e+07
Lakshadweep	1.056446e+06

```
# d. Number of Males vaccinated
total_males_vaccinated = df['Male(Individuals Vaccinated)'].sum()
print(f"\nTotal Males Vaccinated: {total_males_vaccinated:,.0f}")
```

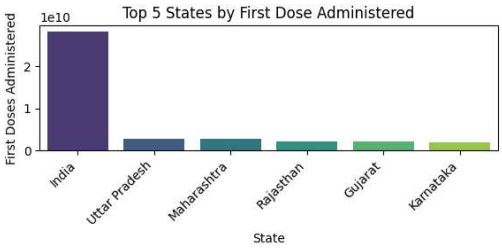
```
Total Males Vaccinated: 7,138,698,858
```

```
# e. Number of Females vaccinated
total_females_vaccinated = df['Female(Individuals Vaccinated)'].sum()
print(f"\nTotal Females Vaccinated: {total_females_vaccinated:,.0f}")
```

```
Total Females Vaccinated: 6,321,628,736
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Visualize top 10 states for First Dose Administered
plt.figure(figsize=(6, 3))
sns.barplot(x=first_dose_statewise.head(6).index, y=first_dose_statewise.head(6).values, palette='viridis')
plt.title('Top 5 States by First Dose Administered')
plt.xlabel('State')
plt.ylabel('First Doses Administered')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```




```
import warnings
warnings.filterwarnings('ignore')
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Visualize top 10 states for Second Dose Administered
plt.figure(figsize=(6, 3))
sns.barplot(x=second_dose_statewise.head(6).index, y=second_dose_statewise.head(6).values, palette='plasma')
plt.title('Top 5 States by Second Dose Administered')
plt.xlabel('State')
plt.ylabel('Second Doses Administered')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
```